Public Health Insights: Challenges in Environment

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Abstract:

The work addresses the intersection between environmental factors and public health by utilizing large datasets containing information on pollutants (like SO2, NO2, and SPM) &healthcare data. Unlike previous research that mainly focuses on health data in isolation, this work emphasizes the importance of considering environmental influences. The primary goal is to predict disease occurrences and conditions based on the combination of these two datasets. The work uses advanced machine learning techniques like the Random Forest Classifier to model the data, ultimately providing visualizations and predictions that help identify vulnerable regions and age groups.

Key words: Environmental factors, Public health, Pollutants (SO2, NO2, SPM), Healthcare data, Machine learning, Random Forest Classifier, Disease preventions, Data modelling, Vulnerable regions, Age groups

Introduction:

Public health, a field concerned with the well-being of a population, is influenced by a complex interplay of factors. These factors include socioeconomic conditions, lifestyle choices, access to healthcare, and environmental conditions. Among these, environmental factors play a particularly significant role in determining the prevalence of diseases, chronic illnesses, and overall health outcomes.

Environmental pollutants, especially in urban areas, have been identified as major contributors to public health problems. Exposure to air pollution, contaminated water, and hazardous substances has been linked to a wide range of health issues, including respiratory diseases (e.g., asthma, bronchitis), cardiovascular conditions (e.g., heart disease, stroke), and cancer. Additionally, environmental pollutants can exacerbate existing health conditions and contribute to a general decline in public health.

This work aims to investigate the relationship between environmental factors and public health outcomes in urban areas. By identifying key environmental pollutants, studying their impact on medical conditions, and utilizing machine learning models to predict health outcomes, this research seeks to provide valuable insights that can inform public health policies and targeted interventions. Ultimately, the goal is to improve the health and well-being of individuals living in polluted environments.

By combining datasets from both healthcare and environmental sources, this work aims to provide predictive insights that can inform public health policies and targeted interventions, especially in areas that are heavily affected by pollution.

Focus of work:

- Combines health and environmental data for predictions.
- Scales to various regions.
- Uses Random Forest model.
- Future use: Load public_health_insights.pkl for new data set.

Literature Survey:

Research on Air Pollution and Health (2010-2020):

- Numerous studies have investigated the association between exposure to air pollutants, such as nitrogen dioxide (NO2) and particulate matter (PM2.5), and respiratory diseases, including asthma and bronchitis. These studies have provided compelling evidence that air pollution is a significant risk factor for these health conditions.
- However, many of these studies have limitations. They often rely on short-term data or focus on smaller geographic areas, making it difficult to generalize their findings to larger populations and diverse environments.

Health Data Analysis (2015-2020):

- Several studies have utilized health records to develop predictive models for disease outcomes based on patient demographics and health history. These models can provide valuable insights into individual health risks.
- Nevertheless, these models have drawbacks. By neglecting environmental factors, such as air pollution, they fail to capture a crucial component of health determinants. This incompleteness limits their ability to accurately predict health outcomes in real-world settings.

Gaps in Previous Research:

- Lack of Integrated Datasets: One of the primary gaps in existing research is the absence of comprehensive datasets that combine health information with environmental data. This integration is essential for understanding the complex interplay between environmental factors and health outcomes.
- Isolation of Health-Based Models: Most health-based models have been developed in isolation from environmental impacts, limiting their ability to account for the significant influence of pollution on disease risk.
- Difficulty in Scaling Models: Previous models often struggle to be scaled to national or international levels, hindering their applicability to broader populations and diverse geographic contexts.

Methodologies:

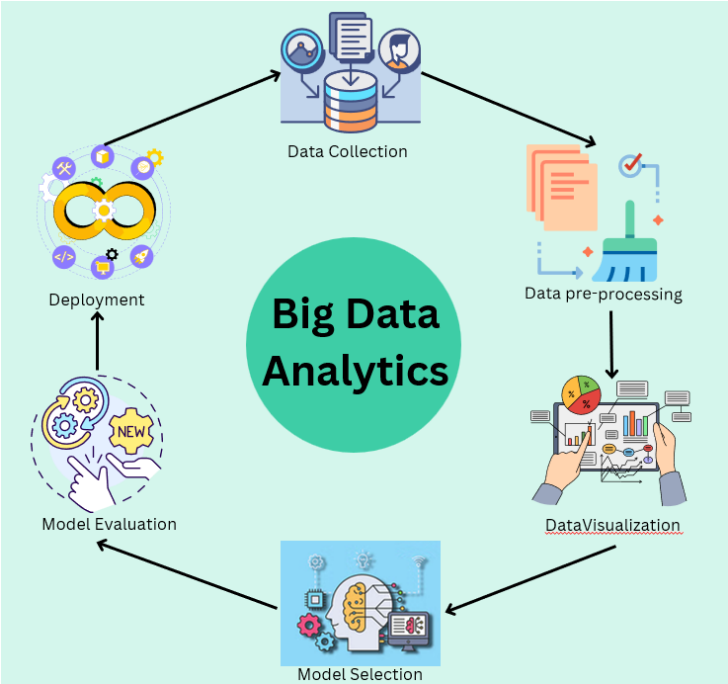


Fig: 4.1 BDA Workflow

- **Data Collection:** This initial phase involves gathering relevant data from various sources, such as electronic health records (EHRs), clinical trials, public health surveys, and government databases. The collected data can be structured, unstructured, or a combination of both.
- **Data Pre-processing:** Once the data is collected, it often requires cleaning, normalization, and transformation to ensure its quality and consistency. This step involves tasks like handling missing values, removing outliers, and converting data into a suitable format for analysis.
- **Big Data Analytics:** The core of the work lies in analyzing the large volume of collected data using advanced analytics techniques. This may include statistical analysis, machine learning algorithms, and data mining methods to extract valuable insights and patterns.
- **Model Selection:** Based on the analysis results, suitable models are selected to address specific research questions or public health challenges. These models can be used for prediction, classification, or other analytical purposes.
- **Model Evaluation:** The selected models are evaluated to assess their performance and accuracy. This involves using appropriate metrics and techniques to measure the models' effectiveness in solving the intended problems.
- **Data Visualization:** The results of the analysis and modelling are presented in a visually appealing and understandable manner. This helps communicate the findings to stakeholders, policymakers, and the public.
- **Deployment:** The final step involves deploying the developed models or insights into real-world applications. This may include integrating the models into existing healthcare systems, developing new tools or applications, or providing recommendations for public health interventions.

Algorithms:

Step 1: Load the dataset

Step 2: Ensure 'date' columns are in the correct format

Step 3: Check for common dates

Step 4: Merge datasets on 'date'

Step 5: Handle missing values

Step 6: Encode target variable (y) if it's categorical

Step 7: Split data into training and testing sets

Step 8: Standardize features

Step 9: Create and train the Random Forest Classifier model

Step 10: Make predictions

- Model can Predict values by using .predict()

Step 11: Evaluate the model

Step 12: Save the model for later use

Based on Algorithm How it works example:

Name	Age	Gender	Blood Type	Medical Condition	date	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
Bobby JacksOn	30	Male	B-	Cancer	01-07-1992	Matthew Smith	Sons and Miller	Blue Cross	18856.28131	328	Urgent	02-02-2024	Paracetamol	Normal
LesLie TErRy	62	Male	A+	Obesity	02-07-1992	Samantha Davies	Kim Inc	Medicare	33643.32729	265	Emergency	26-08-2019	Ibuprofen	Inconclusive
DaNnY sMitH	76	Female	A-	Obesity	03-07-1992	Tiffany Mitchell	Cook PLC	Aetna	27955.09608	205	Emergency	07-10-2022	Aspirin	Normal
andrEw waTtS	28	Female	O+	Diabetes	04-07-1992	Kevin Wells	Hernandez Rogers and Vang,	Medicare	37909.78241	450	Elective	18-12-2020	Ibuprofen	Abnormal
adrIENNE bEll	43	Female	AB+	Cancer	05-07-1992	Kathleen Hanna	White-White	Aetna	14238.31781	458	Urgent	09-10-2022	Penicillin	Abnormal
EMILY JOHNSOn	36	Male	A+	Asthma	06-07-1992	Taylor Newton	Nunez-Humphrey	UnitedHealthcare	48145.11095	389	Urgent	24-12-2023	Ibuprofen	Normal
edwArD EDWaRDs	21	Female	AB-	Diabetes	07-07-1992	Kelly Olson	Group Middleton	Medicare	19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive
ChrisTInA MARtinez	20	Female	A+	Cancer	08-07-1992	Suzanne Thomas	Powell Robinson and Valdez,	Cigna	45820.46272	277	Emergency	07-01-2022	Paracetamol	Inconclusive
JASmINe aGuilaR	82	Male	AB+	Asthma	09-07-1992	Daniel Ferguson	Sons Rich and	Cigna	50119.22279	316	Elective	14-07-2020	Aspirin	Abnormal
ChRISTopher BerG	58	Female	AB-	Cancer	10-07-1992	Heather Day	Padilla-Walker	UnitedHealthcare	19784.63106	249	Elective	22-06-2021	Paracetamol	Inconclusive
mIchElLe daniELs	72	Male	O+	Cancer	11-07-1992	John Duncan	Schaefer-Porter	Medicare	12576.79561	394	Urgent	22-04-2020	Paracetamol	Normal
aaRon MARtiNeZ	38	Female	A-	Hypertension	12-07-1992	Douglas Mayo	Lyons-Blair	Medicare	7999.58688	288	Urgent	05-09-2023	Lipitor	Inconclusive
connOR HANsEn	75	Female	A+	Diabetes	13-07-1992	Kenneth Fletcher	Powers Miller, and Flores	Cigna	43282.28336	134	Emergency	28-12-2019	Penicillin	Abnormal
rObeRt bAuer	68	Female	AB+	Asthma	14-07-1992	Theresa Freeman	Rivera-Gutierrez	UnitedHealthcare	33207.70663	309	Urgent	19-06-2020	Lipitor	Normal
bROOkE brady	44	Female	AB+	Cancer	15-07-1992	Roberta Stewart	Morris-Arellano	UnitedHealthcare	40701.59923	182	Urgent	13-10-2021	Paracetamol	Normal
MS. nAtaliE gAMble	46	Female	AB-	Obesity	16-07-1992	Maria Dougherty	Cline-Williams	Blue Cross	12263.35743	465	Elective	11-01-2023	Aspirin	Inconclusive
haley perkins	63	Female	A+	Arthritis	17-07-1992	Erica Spencer	Cervantes-Wells	UnitedHealthcare	24499.8479	114	Elective	14-07-2020	Paracetamol	Normal
mRS. jamiE cAMPBELl	38	Male	AB-	Obesity	18-07-1992	Justin Kim	Torres, and Harrison Jones	Cigna	17440.46544	449	Urgent	02-04-2020	Paracetamol	Abnormal
LuKE BuRgEss	34	Female	A-	Hypertension	19-07-1992	Justin Moore Jr.	Houston PLC	Blue Cross	18843.02302	260	Elective	14-03-2021	Aspirin	Abnormal
dANIEL schmIdt	63	Male	B+	A						465	Elective	22-11-2022	Penicillin	Normal

Fig 4.2 :health_care dataset

environment.csv

File Origin

1252: Western European (Windows

stn_code	sampling_date	state	location	agency	type	so2	no2	rspm	date
150	February - M021990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	4.8	17.4	180	01-02-1990
151	February - M021990	Andhra Pradesh	Hyderabad	NA	Industrial Area	3.1	7	130	01-02-1990
152	February - M021990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	6.2	28.5	115	01-02-1990
150	March - M031990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	6.3	14.7	100	01-03-1990
151	March - M031990	Andhra Pradesh	Hyderabad	NA	Industrial Area	4.7	7.5	99	01-03-1990
152	March - M031990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	6.4	25.7	90	01-03-1990
150	April - M041990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	5.4	17.1	100	01-04-1990
151	April - M041990	Andhra Pradesh	Hyderabad	NA	Industrial Area	4.7	8.7	101	01-04-1990
152	April - M041990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	4.2	23	100	01-04-1990
151	May - M051990	Andhra Pradesh	Hyderabad	NA	Industrial Area	4	8.9	111	01-05-1990
152	May - M051990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	3.6	18.6	102	01-05-1990
150	June - M061990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	3.9	14.1	133	01-06-1990
151	June - M061990	Andhra Pradesh	Hyderabad	NA	Industrial Area	5.6	11.8	82	01-06-1990
152	June - M061990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	3.3	19.3	111	01-06-1990
150	July - M071990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	3.9	8.2	118	01-07-1990
152	July - M071990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	3.5	12.1	135	01-07-1990
151	July - M071990	Andhra Pradesh	Hyderabad	NA	Industrial Area	7.9	10.2	80	01-07-1990
150	August - M081990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	4	9.9	179	01-08-1990
151	August - M081990	Andhra Pradesh	Hyderabad	NA	Industrial Area	12.4	11.5	58	01-08-1990
152	August - M081990	Andhra Pradesh	Hyderabad	NA	Residential, Rural and other Areas	4	12.3	99	01-08-1990

Fig 4.3: environment data set

The environmental & healthcare dataset should follow the correct date format: dd-mm-yy

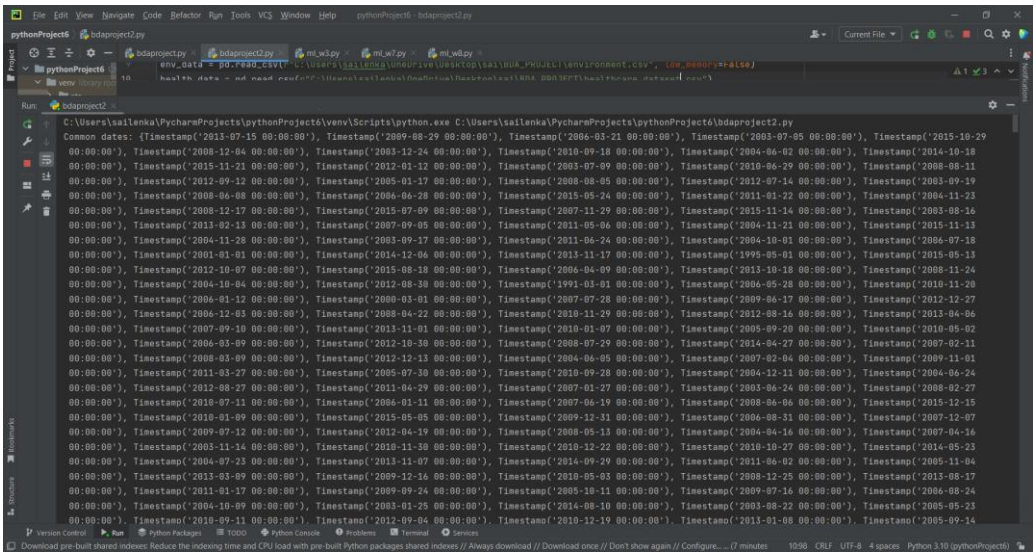


Fig 4.4: common dates output

```
Shape of merged_data: (1307451, 24)
  stn_code       sampling_date  ...  Medication Test Results
0      150  February - M021990  ...  Penicillin     Abnormal
1      150  February - M021990  ...     Lipitor       Normal
2      150  February - M021990  ...     Aspirin     Abnormal
3      150  February - M021990  ...     Lipitor       Normal
4      151  February - M021990  ...  Penicillin     Abnormal

[5 rows x 24 columns]
Columns in merged_data: Index(['stn_code', 'sampling_date', 'state', 'location', 'agency', 'type',
       'so2', 'no2', 'spm', 'date', 'Name', 'Age', 'Gender', 'Blood Type',
       'Medical Condition', 'Doctor', 'Hospital', 'Insurance Provider',
       'Billing Amount', 'Room Number', 'Admission Type', 'Discharge Date',
       'Medication', 'Test Results'],
```

Fig 4.5: merged data and top 5 rows shows

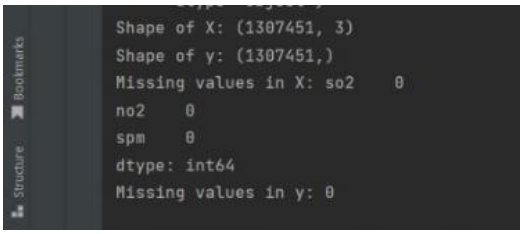


Fig 4.6: missing values handled

Random Forest Classifier:

A supervised learning algorithm that builds multiple decision trees and merges them to produce more accurate predictions. The classifier is well-suited for handling large datasets with multiple features. The Random Forest classifier is an ensemble learning method that operates by constructing a multiple of decision trees during training and outputting the class that is the mode of the classes (classification) of the individual trees.

- **Training**: The data will be first pre-processed. The data is split into training and testing sets. The model is trained using the Random Forest algorithm to predict medical conditions based on environmental factors.
- **Evaluation**: Evaluating a Random Forest Classifier typically involves assessing how well the model generalizes to unseen data. Performance metrics like accuracy, precision, recall, and F1 score are used to evaluate the classifier.

- **Accuracy:** The proportion of correctly predicted instances out of the total instances.

$$Accuracy = \frac{\text{No. of correct predictions}}{\text{Total no. of predictions}}$$

- **Precision**: The proportion of positive predictions that are actually correct.

$$Precision = \frac{\text{True positives}}{\text{True positives} + \text{False positives}}$$

- **Recall:** The proportion of actual positives that are correctly predicted.

$$Recall = \frac{\text{True positives}}{\text{True positives} + \text{False negatives}}$$

- **F1 score:** The harmonic mean of precision and recall, balancing the two.

$$F1\ Score = 2 \times \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

```
Model saved as 'public_health_insights_classifier_model.pkl'
```

Fig 4.7: save the model for future

Results and Discussions:

Visualizations of environment data set:

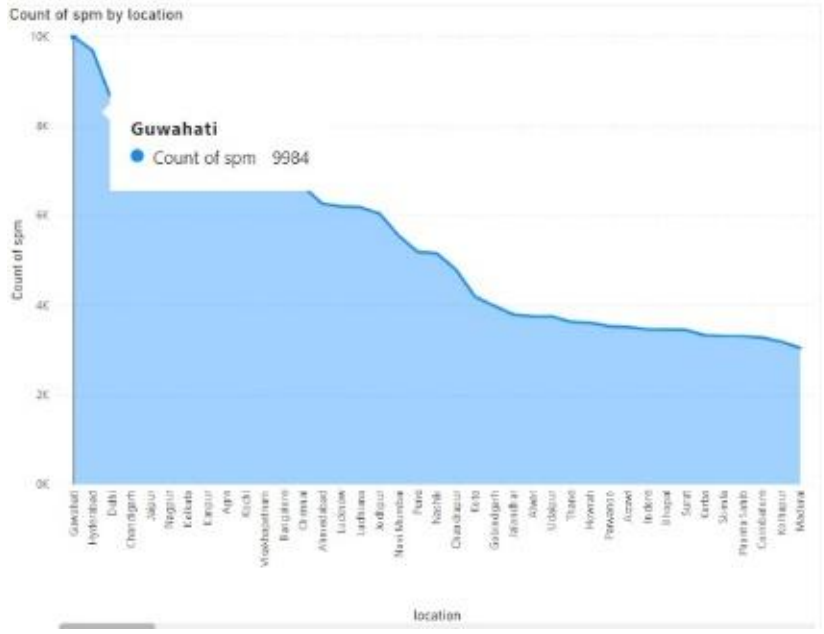


Fig 5.1: count of SPM by location

The visualization is a bar chart that shows the count of SPM (suspended particulate matter) for different locations. The x-axis represents the locations, and the y-axis represents the count of SPM. The chart is sorted in descending order of the count of SPM. The highest count of SPM is for Guwahati, followed by Hyderabad and Chandigem. The chart also shows the count of SPM for Visakhapatnam, which is the current location.

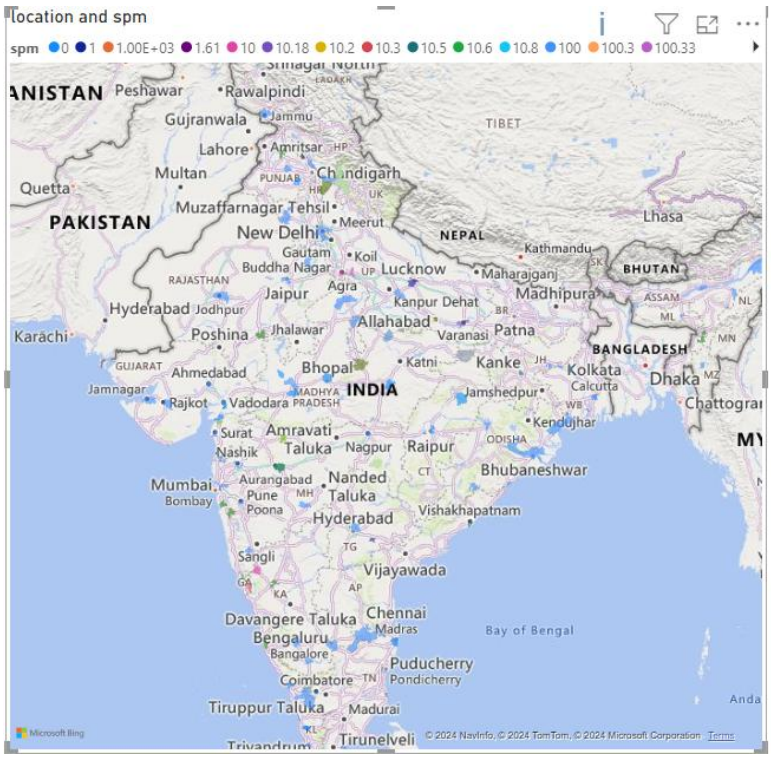


Fig 5.2:map visualization for SPM based on location

The visualization is a map of India with different locations marked with varying colors. The colors represent the level of SPM (suspended particulate matter) in those locations. The darker the color, the higher the level of SPM. The current location, Visakhapatnam, is marked with a blue circle. The map also shows the neighboring countries of India, including Pakistan, Nepal, Bhutan, Bangladesh, and Myanmar.

Visualizations of healthcare data set:

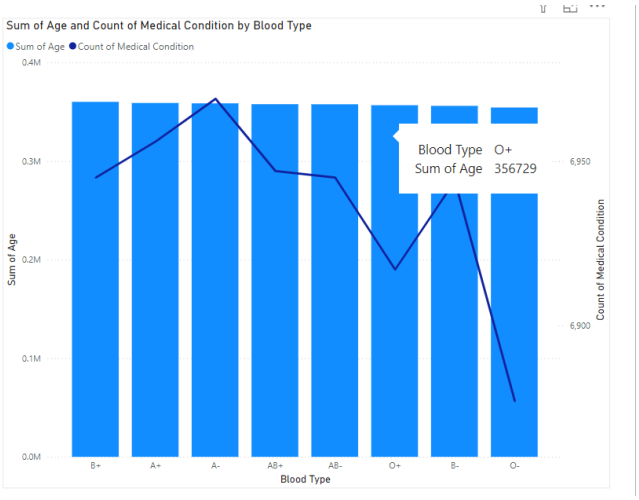


Fig 5.3:based on age and blood type medical condition

The visualization is a combination chart that shows the sum of age and the count of medical conditions for different blood types. The x-axis represents the blood types, and the y-axis represents the sum of age and the count of medical conditions. The chart shows two lines: one for the sum of age and one for the count of medical conditions. The sum of age is highest for blood

type O+, followed by B+, A+, and AB+. The count of medical conditions is highest for blood type B+, followed by O+, A+, and AB+.

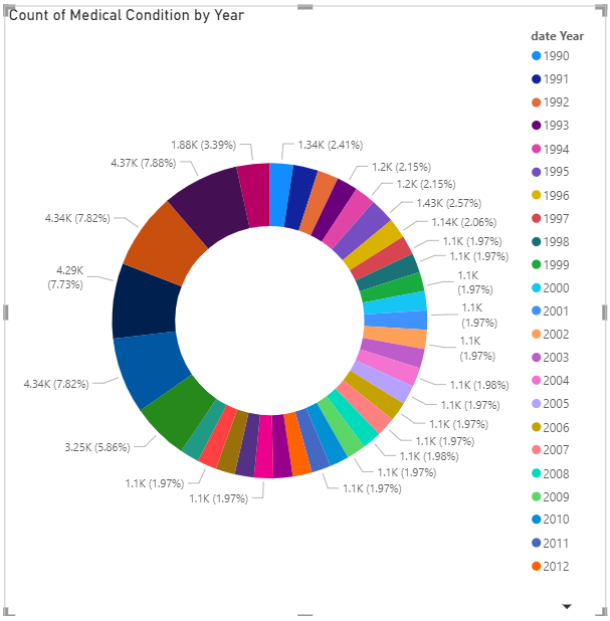


Fig 5.4: medical condition on year

The visualization is a pie chart that shows the count of medical conditions for different years. The chart is divided into segments, each representing a year. The size of each segment corresponds to the count of medical conditions for that year. The chart also shows the percentage of the total count of medical conditions for each year. The year with the highest count of medical conditions is 1990, followed by 2004 and 1994. The chart also shows the current year, 2012, which has a count of medical conditions of 1.1K (1.97%).

Model Performance:

- The Random Forest model produced high accuracy scores, indicating its robustness for predicting health conditions based on environmental factors.
- **Precision** and **recall** values were also high, suggesting that the model effectively distinguished between different medical conditions.
- **Support:** The total number of true positives (actual positives) and true negatives (actual negatives) for each class. This is particularly relevant for imbalanced datasets, where some classes might have fewer examples.
- The **F1-score** handled the imbalance between age groups and regions, making it suitable for skewed datasets.
- **Macro and Weighted Averages:**

  Macro Average: Calculates the unweighted mean of precision, recall, and F1-score across all classes. This treats each class equally regardless of its size.

  Weighted Average: Takes into account class imbalances by assigning more weight to larger classes. This provides a better overall picture of performance when dealing with skewed datasets.

Conclusion and Extension:

The work successfully demonstrated how integrating health and environmental data can lead to more accurate and actionable insights in public health. The Random Forest model provided strong predictions, and the visualizations offered a clear picture of pollution's impact on health. Future work can involve adding real-time environmental data feeds, incorporating other pollutants like PM2.5 and PM10, and extending the model to predict more types of medical conditions.

Future Extensions:

- Expanding the model to global datasets.
- Incorporating additional machine learning algorithms like deep learning or neural networks for more complex predictions for more accuracy.

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